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BRAC UNIVERSITY
Merul Badda, Dhaka, Bangladesh
CSE331 : Automata and Computability
Quiz 1
Summer 2024

Duration: 30 minutes

CO1

Total: 20 marks

Name: Radia Rahman	ID: 22299495	Section: 21
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$L_1 = \{w \in \Sigma^* \mid w \text{ starts with } 10\}$, where $\Sigma = \{0, 1\}$
 $L_2 = \{w \in \Sigma^* \mid w \text{ contains an even number of } 0\text{'s}\}$, where $\Sigma = \{0, 1\}$

DFA

1. Draw the state diagram of the DFA for the following regular languages: (5 x 2 = 10 marks)

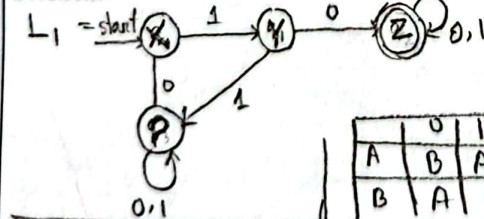
$L(M) \rightarrow L_2$

Solution:



$L(M) \rightarrow L_1 \cap L_2$

Solution:



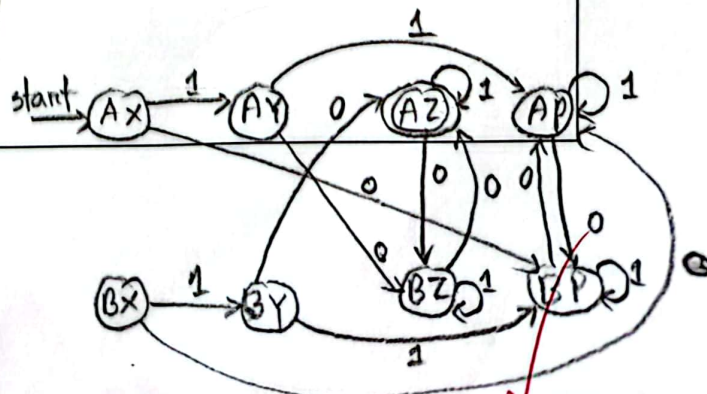
	x	y	z	p
A	AX	AY	AZ	AP
B	BX	BY	BZ	BP

	0	1
A	B	A
B	A	B

	0	1
X	P	Y
Y	Z	P
Z	Z	Z
P	P	P

$(A, 0) = B, (X, 0) = P \mid AX, 0 = BP$
 $(Y, 0) = Z \mid AY, 0 = BZ$
 $(Z, 0) = Z \mid AZ, 0 = BZ$
 $(P, 0) = P \mid AP, 0 = BP$

$(A, 1) = A, (X, 1) = Y \mid AX, 1 = AY$
 $(Y, 1) = P \mid AY, 1 = AP$
 $(Z, 1) = Z \mid AZ, 1 = AZ$
 $(P, 1) = P \mid AP, 1 = AP$

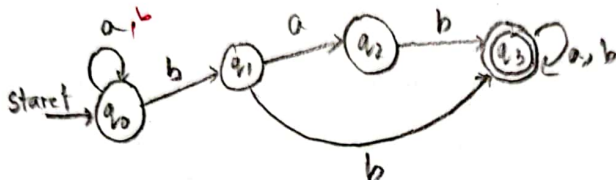


NFA

2. Construct the NFA for the following regular languages: (5 marks)

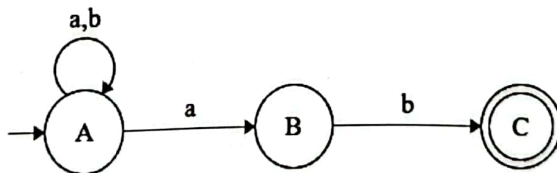
$L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains } bab \text{ or } bb\}$, where $\Sigma = \{a, b\}$ using only 4 states.

Solution:



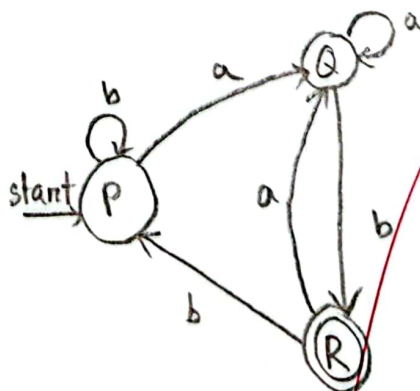
NFA → DFA

3. Convert the following NFA into equivalent DFA:



Solution: Table for DFA:

state \	a	b
P = {A}	{A, B} (Q)	{A} (P)
Q = {A, B}	{A, B} (Q)	{A, C} (R)
R = {A, C}	{A, B} (Q)	{A} (P)



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Name: <u>NAKSHATRA Roy</u>	ID: <u>22299245</u>	Section: <u>19</u>
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$L_1 = \{w \in \Sigma^* \mid w \text{ starts with } 10\}$, where $\Sigma = \{0, 1\}$
 $L_2 = \{w \in \Sigma^* \mid w \text{ contains an odd number of } 1\text{'s}\}$, where $\Sigma = \{0, 1\}$

DFA

1. Draw the state diagram of the DFA for the following regular languages: (5 x 2 = 10 marks)

$L(M) \rightarrow L_2$
Solution:

$L(M) \rightarrow L_1 \cup L_2$
Solution:

$L(M) \rightarrow L_1$

$L(M) \rightarrow L_2$

	x	y
A	Ax	Ay
B	Bx	By
C	Cx	Cy
D	Dx	Dy

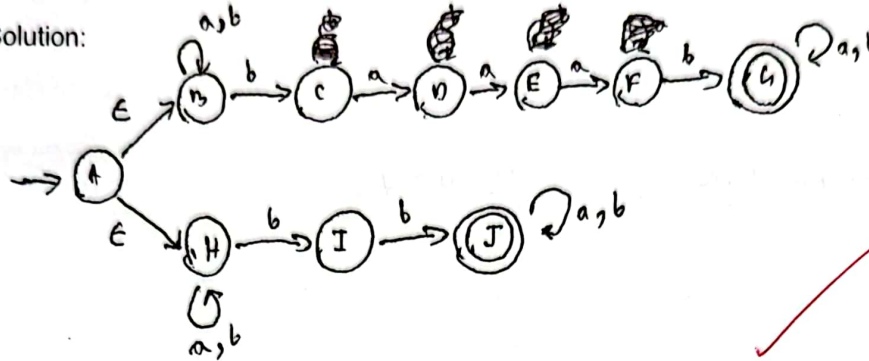
if $w = 010$ ← odd no. of 1s
still accepted
so correct

NFA

2. Construct the NFA for the following regular languages: (5 marks)

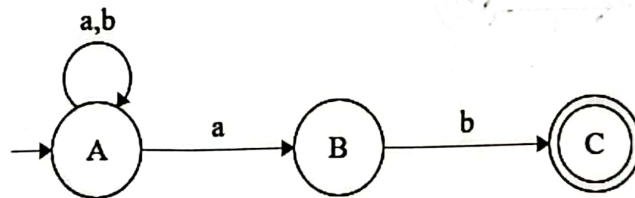
$L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains baaab or bb}\}$, where $\Sigma = \{a, b\}$.

Solution:



NFA $\rightarrow$ DFA

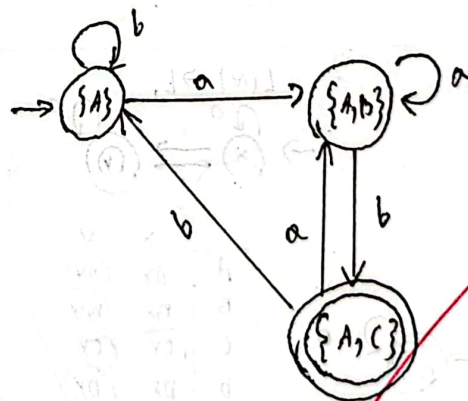
3. Convert the following NFA into equivalent DFA:



Solution:

state	a	b
$\{A\}$	$\{A, B\}$	$\{A\}$
$\{A, B\}$	$\{A, B\}$	$\{A, C\}$
$\{A, C\}$	$\{A, B\}$	$\{A\}$

end state
DFA table



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Quiz 2
Spring 2024

Duration: 35 minutes

CO2

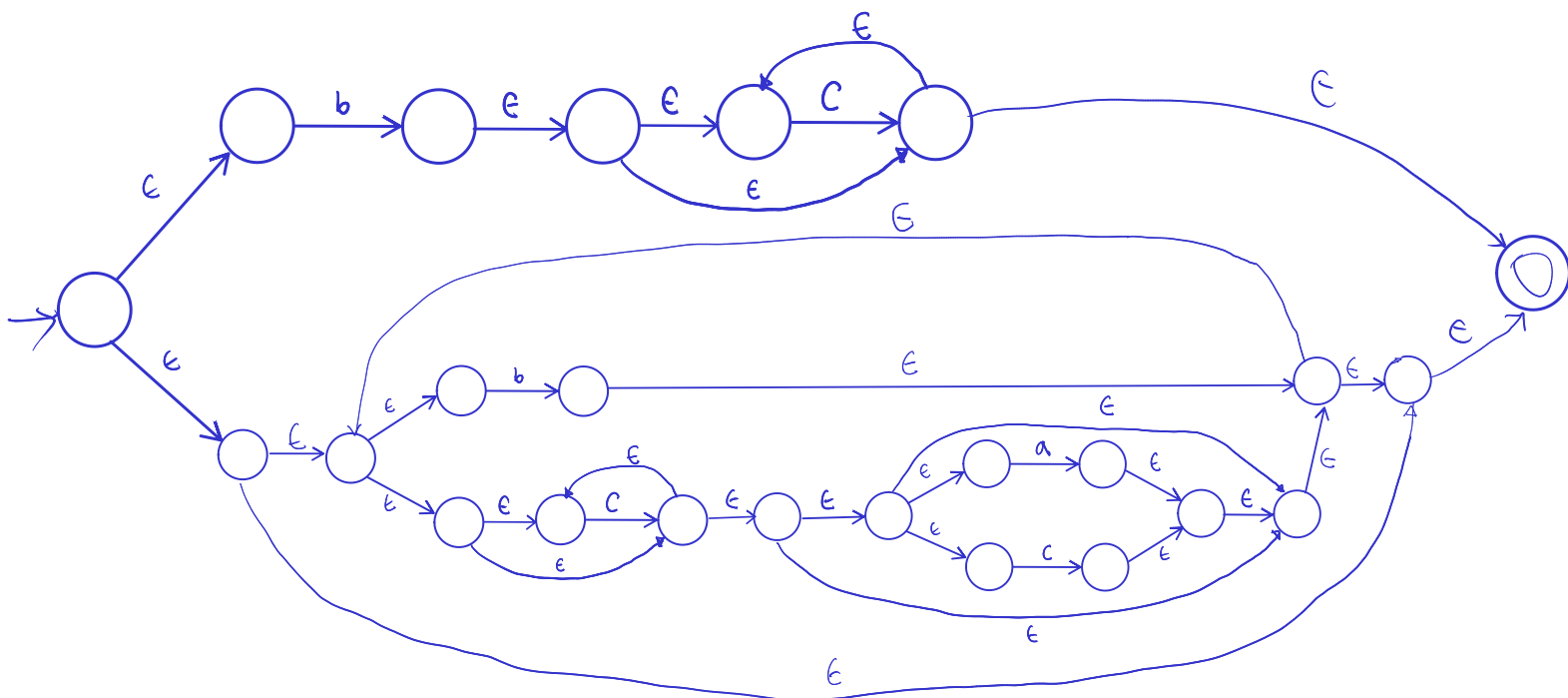
Total: 20 marks

Name:	ID:	Section: 19
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RE TO NFA [6 points]

Convert the following regular expression over $\Sigma = \{a, b, c\}$ into an equivalent NFA.

$$bc^* \cup (b \cup c^* (a \cup c)^*)^*$$



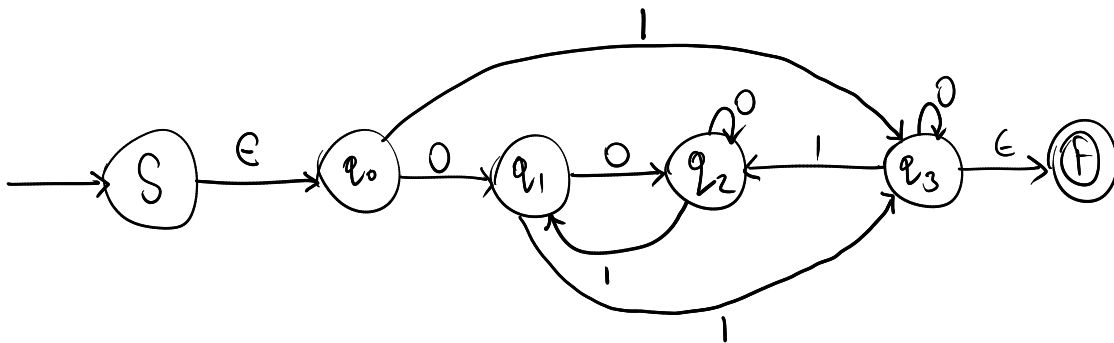
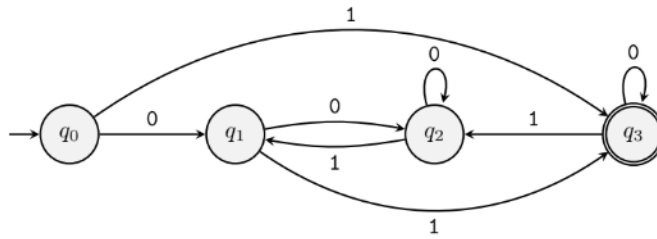
Conditions

The NFA derived from Thompson's Construction must follow some conditions:

- ✓ 1. There can be only one start state.
- ✓ 2. The start state cannot be accessible from any states i.e. it cannot have transitions to it from other states.
- ✓ 3. There can be only one final state. The final state cannot have any transition from it to other states.
- ✓ 4. The number of transitions leaving any state is at most two.

DFA to RE [6 points]

Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_1 , then q_2 and finally q_3 . You must show work.

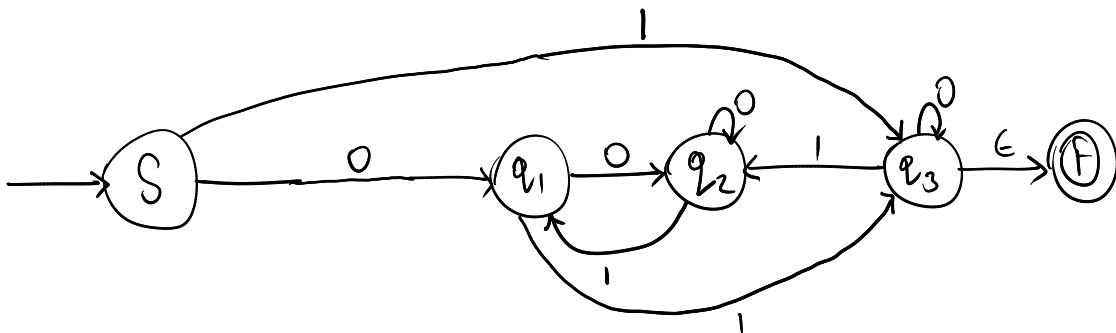


Eliminate q_0

In	out
S	q_3
	q_1

$$S \xrightarrow{1} q_3$$

$$S \xrightarrow{0} q_1$$



Eliminate q_1

In	out
S	q_2
q_2	q_3

$$S \xrightarrow{00} q_2$$

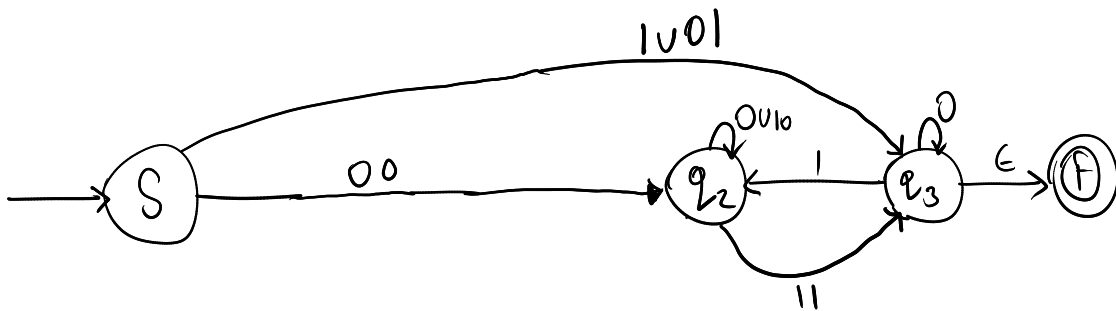
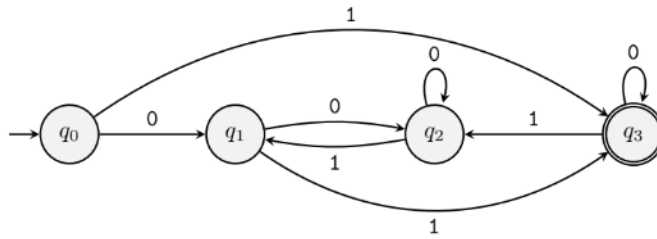
$$S \xrightarrow{01} q_3$$

$$q_2 \xrightarrow{10} q_2$$

$$q_2 \xrightarrow{11} q_3$$

DFA to RE [6 points]

Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_1 , then and finally q_3 . You must show work.

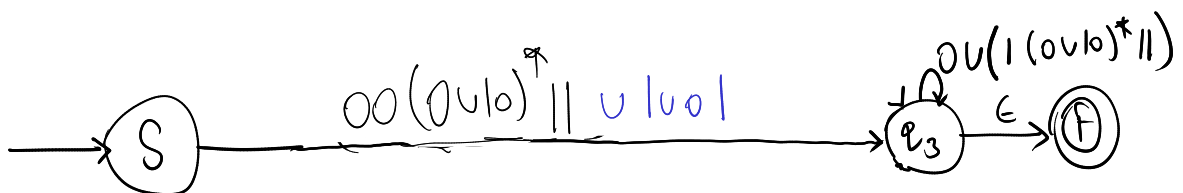


Eliminate q_2

In	out
S	q_3
q_3	

$$S \xrightarrow{00(0010)^*11} q_3$$

$$q_3 \xrightarrow{1(0010)^*11} q_3$$



Eliminate q_3

In	out
S	F

$$\text{regex: } 00(0010)^*11 \cup (1001)(00(1001)^*11)^*$$

Regular expression [3 + 2 + 3 points]

Let $\Sigma = \{0, 1\}$. Give regular expressions generating each of the following languages over Σ .

- a) $\{w : \text{every } 1 \text{ in } w \text{ is followed by an even number of } 0\text{s}\}$
- b) $\{w : w \text{ does not contain } 10\}$
- c) $\{w : 10 \text{ appears in } w \text{ exactly once}\}$
(Hint: If $w = x10y$, what can you say about x and y ?)

(a) $0^* (1 (00)^*)^*$

(b) $0^* 1^*$

(c) $0^* 1^* 10 0^* 1^*$

Take a look at the grammar below and solve the following problems.

$$S \rightarrow AB \mid bAA \mid SS$$

$$A \rightarrow a \mid aA$$

$$B \rightarrow b \mid \epsilon$$

- Give a leftmost derivation for the string abaaab. (3 points)
- Sketch the parse tree corresponding to the derivation you gave in (a). (2 points)
- Demonstrate that the given grammar is ambiguous by showing two more parse tree (apart from the one you already found in (b)) for the same string. (4 points)
- Find a string w of length five such that w has exactly one parse tree in the grammar above. (1 point)

a, b)

Leftmost derivation	Corresponding Parse Tree
$ \begin{aligned} S &\rightarrow SS \rightarrow SSS \rightarrow ABSS \rightarrow aBSS \rightarrow \\ &abSS \rightarrow abABS \rightarrow abaBS \rightarrow aba\epsilon S \\ &\rightarrow abaSS \rightarrow abaABS \rightarrow abaabS \rightarrow \\ &abaa\epsilon S \rightarrow abaaAB \rightarrow abaaaB \rightarrow abaaab \end{aligned} $	<pre> graph TD S1[S] --- S2[S] S1 --- S3[S] S2 --- S4[S] S2 --- S5[S] S3 --- S6[S] S3 --- S7[S] S4 --- A1[A] S4 --- B1[B] S5 --- A2[A] S5 --- B2[B] S6 --- A3[A] S6 --- B3[B] S7 --- A4[A] S7 --- B4[B] A1 --- a1[a] B1 --- b1[b] A2 --- a2[a] B2 --- e1["ε"] A3 --- a3[a] B3 --- e2["ε"] A4 --- a4[a] B4 --- b2[b] </pre>

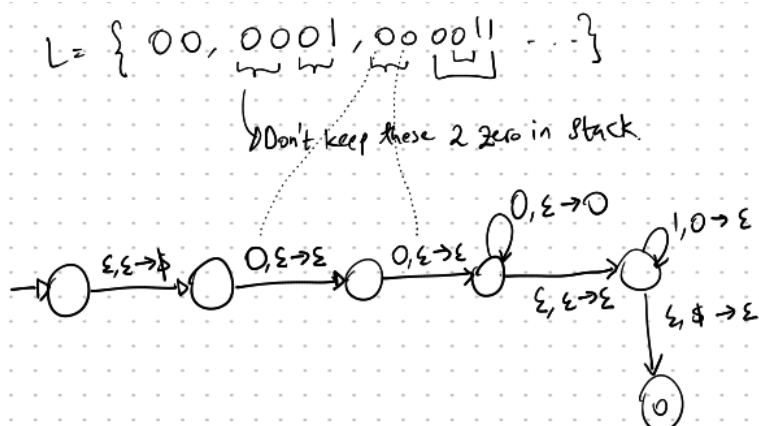
c)

$ \begin{aligned} S &\rightarrow SS \rightarrow ABS \rightarrow aBS \rightarrow abS \rightarrow \\ abAB &\rightarrow abaAB \rightarrow abaaAB \rightarrow \\ abaaaB &\rightarrow abaaaab \end{aligned} $	<pre> graph TD S1[S] --- S2[S] S1 --- S3[S] S2 --- A1[A] S2 --- B1[B] A1 --- a1[a] B1 --- b1[b] S3 --- A2[A] S3 --- B2[B] A2 --- a2[a] A2 --- A3[A] A3 --- a3[a] A3 --- A4[A] A4 --- a4[a] B2 --- b2[b] </pre>
$ \begin{aligned} S &\rightarrow SS \rightarrow ABS \rightarrow aBS \rightarrow abS \rightarrow \\ abSS &\rightarrow abABS \rightarrow abaBS \rightarrow \\ abaES &\rightarrow abaAB \rightarrow abaaAB \rightarrow \\ abaaaB &\rightarrow abaaaab \end{aligned} $	<pre> graph TD S1[S] --- S2[S] S1 --- S3[S] S2 --- A1[A] S2 --- B1[B] A1 --- a1[a] B1 --- b1[b] S3 --- S4[S] S3 --- S5[S] S4 --- A2[A] S4 --- B2[B] A2 --- a2[a] B2 --- e1[ε] S5 --- A3[A] S5 --- B3[B] A3 --- a3[a] A3 --- A4[A] A4 --- a4[a] B3 --- b3[b] </pre>

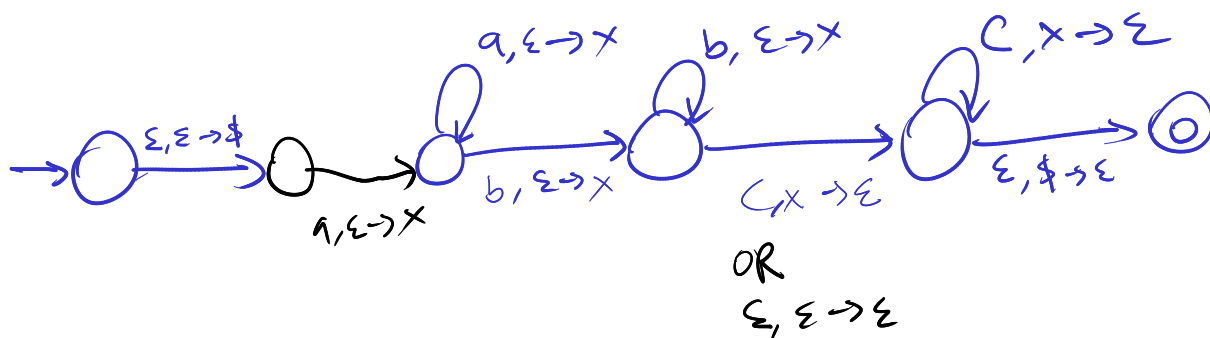
d) abbaa, baaab

Constructing Pushdown Automata

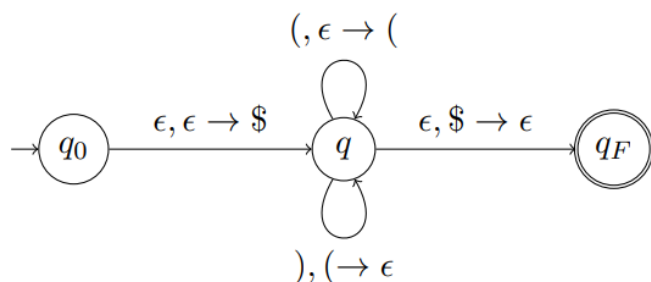
$$L = \{w \in \{0,1\}^* \mid w = 0^{n+2}1^n, n \geq 0\}$$



$$L = \{a^i b^j c^k \mid i, j, k \geq 1, i + j = k\}$$



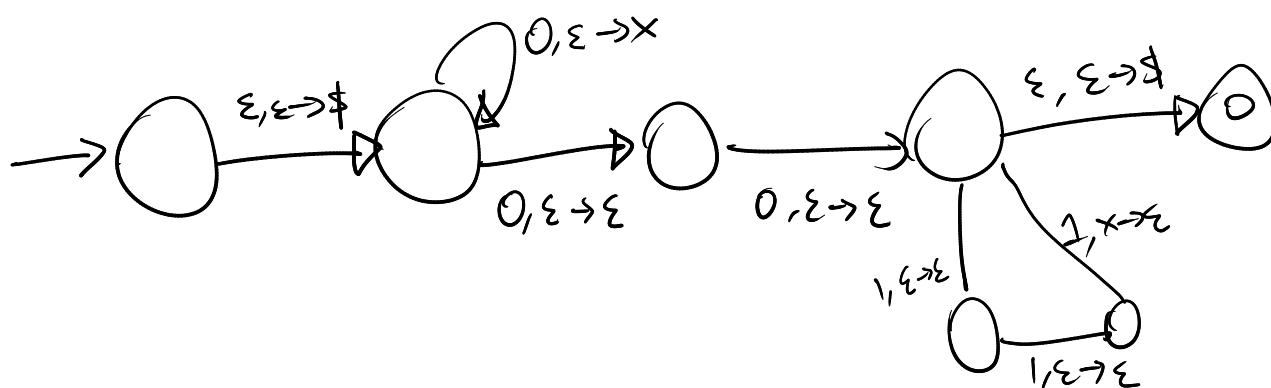
$$L = \{w \in \{ (,) \}^* : w \text{ is a valid parenthesis}\}$$



$$L = \{w \in \{0,1\}^* : 0^{n+2}1^{3n}, \text{ where } n \geq 0\}$$

$$0^{n+2}1^{3n} = 0^n 0^2 1^{3n}$$

$$L = \{00, 000111, 0000111111, \dots\}$$



Matching Parenthesis: PDA execution

